

Stellar PSFs Misassign Light in Little Red Dots

Broadband point-spread functions from field stars can fake extended hosts in compact high-redshift sources.

arXiv:2608.30459 — source-conditioned polychromatic PSFs for little red dots.

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Background

Compact high-redshift sources are often decomposed using field-star point-spread functions. This assumes the source and stars share the same spectral shape across a broad filter.

No test has checked whether spectrally different compact nuclei bias host measurements from stellar PSF fitting.

What this paper is selling

The authors show that mismatched spectra systematically misassign light, creating or inflating apparent extended components. They derive the error's origin and demonstrate a correction for other compact distant sources.

Key results to remember

- A point-only LRD yields wavelength-dependent residuals mistaken for a Sérsic host.
- The spurious F444W host allocation exceeds the published 11.8% component in every test.
- Inferred sizes, luminosities, and masses of LRDs are therefore systematically biased.

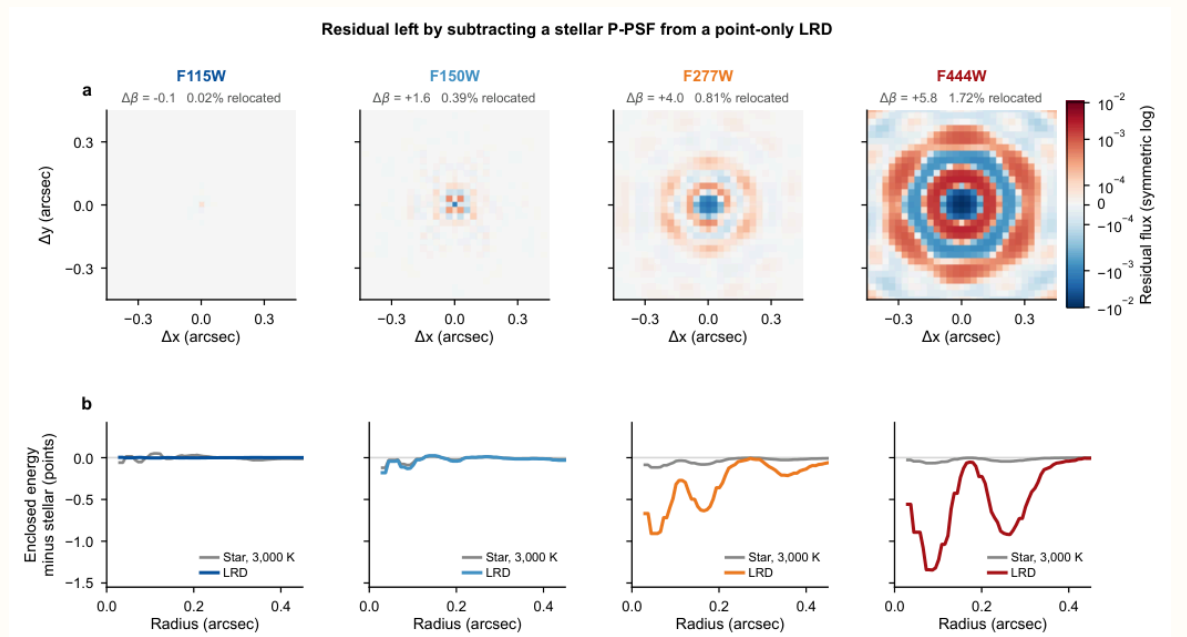


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves filter-dependent residuals that mimic a Sérsic host.

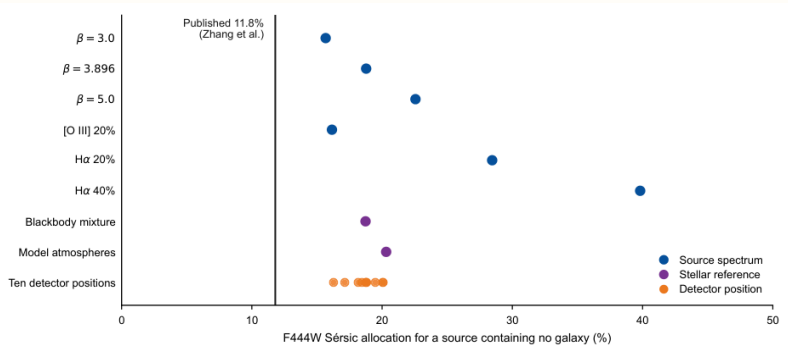


Figure 2 | Every tested nuclear spectrum, stellar reference, and detector position produces a spurious F444W host above 11.8%.